[CLIENT ORGANISATION] | SANITISED SAMPLE DOCUMENT

Business & Software Requirements Specification

Hospital Data Platform - BRD (Part A) and SRS (Part B)

Prepared to IEEE 29148-style structure; all names, systems and data are placeholders.

DOCUMENT ID	DOC-BRD-SRS-001
VERSION	1.0 (aligned to platform v1.1)
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STATUS	Sample - not an approved baseline
CLASSIFICATION	Public sample / synthetic data

Document control

Item	Detail
Document	Business Requirements Document (Part A) and Software Requirements Specification (Part B)
System	Hospital Data Platform (bronze / silver / gold analytics pipeline)
Organisation	[Client Organisation] - a multi-branch hospital group (fictional in this sample)
Sponsor / owner	[Executive Sponsor], [Head of Data and Analytics] - roles only, no named individuals
Sanitisation	No real patient, staff, branch-system or vendor names. Systems appear as generic labels (HIS, Billing System). Email domains use *.example*.

Revision history

Version	Date	Author	Change
0.1	[date]	[Business Analyst]	Initial outline from stakeholder workshops
0.5	[date]	[Business Analyst], [Data Engineer]	Requirements elaborated; traceability added
0.9	13 Oct 2026	[Data Engineer]	Aligned to the implemented platform; verification status added
1.0	13 Oct 2026	[Data Engineer]	Added reconciliation, volume and freshness alerts, tunable thresholds, notification and physician-rate reporting rules (BRQ-17 to 21, FR-045, FR-090 to 105, NFR-11 to 13); closed OI-7; platform counts updated to 104 tests / 12 mutations

Approvals

Role	Name	Signature	Date
Executive Sponsor	[name]		
Head of Data and Analytics	[name]		
Clinical Quality Lead	[name]		
Finance Controller	[name]		
Information Security Officer	[name]		

Template guidance
Items in **[square brackets]** are placeholders to replace. Grey 'Template guidance' boxes explain what belongs in each section and should be deleted from the final baseline. Requirement IDs are stable: never reuse an ID; mark a withdrawn requirement as *Deleted* instead.

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Part A - Business Requirements Document

A1. Purpose and background

[Client Organisation] operates five hospital branches, each with its own Hospital Information System (HIS), plus a central billing system, a physician master and a security matrix. Leadership cannot currently obtain a single trusted count of admissions, a consistent 30-day readmission rate, an explanation of why finance and operations report different monthly revenue, or confidence that users see only the data they are entitled to.

This document states **what the business needs** from a Hospital Data Platform and how success will be judged. Part B translates these needs into verifiable software requirements.

> **Template guidance**
> State the problem in business language. Avoid naming technologies in Part A; technology choices belong in the SRS or design documents.

A2. Business objectives

ID	Objective	Success measure	Target
BO-1	One trusted definition of an inpatient admission	Duplicate / test / cancelled / deleted encounters excluded; every admission accounted for	100% of source admissions reconcile to a flagged or valid row
BO-2	A reliable 30-day unplanned readmission KPI	Rate reproducible at group, branch, department and physician level from one definition	Four grains reconcile to identical totals
BO-3	Revenue that finance and operations can reconcile	Bridge between service-period and transaction-period revenue	Bridge sums exactly to service-period revenue each month
BO-4	Controlled access to patient-level data	Per-user visibility matches entitlement; revoked access ends on next load	0 unauthorised rows visible in test
BO-5	Trustworthy, auditable data quality	Every data defect detected, classified and traceable	100% recall and precision on seeded defects
BO-6	Dependable unattended operation	Safe re-run after failure; clear health signal	Re-run yields identical results; alerts view empty when healthy
BO-7	Early warning when data is incomplete, stale or unreliable	Reconciliation to the source, volume and freshness checks, visible uncertainty on small samples	Every alert type fires on an injected fault; alerts reach people without anyone polling a view

A3. Scope

In scope	Out of scope (this release)
Ingestion of five extracts: admissions, physician assignments, billing, access matrix, change feed	Direct connectivity to live HIS or billing databases
Trusted admission layer with data-quality flags and exception register	Clinical coding (ICD / DRG) and case-mix adjustment
30-day unplanned readmission KPI at four grains	Predictive models and risk scores
Revenue reconciliation (service vs transaction period)	General ledger posting and VAT computation
Row-level security data model driven by an access matrix	Identity provider integration and BI semantic-model build
Operational monitoring, run logging, alert notification and a recovery runbook	24x7 on-call support model

A4. Stakeholders

Role	Interest	Involvement
Executive Sponsor	Group performance and quality reporting	Approves scope and budget
Clinical Quality Lead	Readmission definition and exclusions	Signs off clinical rules BR-4 to BR-7
Finance Controller	Revenue figures and reconciliation	Signs off revenue rules BR-2 and BR-8
Branch Directors	Branch-level KPIs and data access	Consumers; validate branch views
Data Stewards	Exception triage and master-data fixes	Act on exception register
Information Security Officer	Access control and audit	Signs off the security requirements
Data Engineering	Build, test and operate the platform	Delivery team

A5. Business requirements

Priority uses MoSCoW: **M**ust, **S**hould, **C**ould, **W**on't (this release).

ID	Business requirement	Obj.	Pri.
BRQ-01	The business shall have one definition of a valid admission that excludes cancelled, test and source-deleted encounters and uses only the latest version of each encounter.	BO-1	M
BRQ-02	No source admission shall be silently dropped; impossible or incomplete records shall remain visible and be clearly marked.	BO-1, BO-5	M
BRQ-03	Activity shall be attributed to the physician assignment in effect on the event date, not the physician's current assignment.	BO-1, BO-2	M
BRQ-04	The business shall measure the 30-day unplanned readmission rate. Planned returns, deaths and transfers shall not count as readmissions, and patients without a full 30-day observation window shall be excluded from the denominator.	BO-2	M
BRQ-05	The readmission rate shall be reportable by group, branch, department and physician from a single definition.	BO-2	M
BRQ-06	Revenue shall be reportable by service period and by transaction period, with credit notes and adjustments retaining their sign, and a bridge explaining the difference.	BO-3	M
BRQ-07	Invoices not linked to a known admission shall remain in revenue but be flagged for reconciliation.	BO-3, BO-5	M
BRQ-08	Users shall see only the branches and departments they are entitled to; inactive or revoked entitlements shall grant nothing.	BO-4	M
BRQ-09	All data-quality exceptions shall be recorded with severity and treatment, retained per run so that quality can be trended.	BO-5	M
BRQ-10	Late-arriving corrections and source-side deletions shall be incorporated correctly without double counting.	BO-1, BO-6	M
BRQ-11	A change in a source file's structure shall not corrupt data silently; breaking changes shall halt only the affected feed.	BO-6	M
BRQ-12	The pipeline shall be safely re-runnable, and operators shall have a single check showing whether anything is wrong.	BO-6	M
BRQ-13	Data stewards shall receive a formatted exception register they can work from.	BO-5	S
BRQ-14	The platform should scale to group volumes of several hundred million rows by using a production warehouse design.	BO-6	S
BRQ-15	Business users could be offered a dashboard summarising KPIs and exceptions.	BO-2	C
BRQ-16	Real-time (sub-minute) data availability.	-	W

ID	Business requirement	Obj.	Pri.
BRQ-17	The business shall be told when the data loaded differs from the data the source sent, including when row counts agree but amounts do not.	BO-5, BO-7	M
BRQ-18	Physician-level readmission rates shall not be presented as reliable performance when based on small numbers; the uncertainty and a minimum-volume rule shall be visible.	BO-2, BO-7	M
BRQ-19	Operators should be notified of alert conditions without having to poll a view, and a notification outage shall not affect the data pipeline.	BO-7	S
BRQ-20	Alert and reporting thresholds should be adjustable by operators without a software release.	BO-6, BO-7	S
BRQ-21	A truncated or silent feed shall be detected even when its structure is valid.	BO-7	M

A6. Assumptions, constraints and dependencies

Type	Statement
Assumption	Source teams provide extracts with the agreed columns, a version indicator and a modification timestamp.
Assumption	A change feed supplies, for each change, the time it became available to the data platform.
Assumption	Physician assignment dates are inclusive calendar dates.
Constraint	Patient and staff data in non-production environments must be synthetic.
Constraint	Business logic must be reviewable as plain SQL by clinical and finance stakeholders.
Dependency	Clinical Quality Lead confirms the readmission interpretation (planned-return handling; transfers).
Dependency	Information Security confirms masking, encryption and audit controls for any production deployment.

A7. Business risks

Risk	Impact	Mitigation (requirement)
Source system changes unannounced	Failed or silently wrong loads	BRQ-11, BRQ-12
Clinical definition interpreted differently by stakeholders	KPI disputes	BRQ-04, sign-off in A4; rule isolated for easy change
Stale access grants	Over-exposure of patient data	BRQ-08
Late corrections or deletes missed	Overstated volumes or revenue	BRQ-10
Truncated or partial extract with valid structure	Understated volumes; wrong KPIs	BRQ-17, BRQ-21
Small-sample rates over-interpreted	Unfair or misleading provider comparison	BRQ-18

A8. Acceptance of the business case

The business accepts the release when every **Must** requirement is verified (Part B, Section B9), the Clinical Quality Lead has signed off BR-4 to BR-7, the Finance Controller has signed off BR-2 and BR-8, and the Information Security Officer has accepted the access model.

Part B - Software Requirements Specification

B1. Introduction

Purpose. This part specifies the functional, data, interface and quality requirements of the Hospital Data Platform so that they can be designed, built and verified. **Audience:** data engineers, testers, data stewards and reviewers. **Convention:** 'shall' denotes a mandatory requirement; 'should' a recommended one.

B2. Overall description

B2.1 Product perspective

The platform sits between source systems and reporting tools. It lands source extracts unchanged (bronze), produces trusted flagged records (silver), publishes business-ready facts, dimensions and KPI views (gold), and maintains a control layer (watermark, run log, exception register, schema snapshot) that observes every step.

B2.2 User classes

User class	Needs from the system
Analyst / report author	Query conformed facts, dimensions and KPI views
Branch or department head	See only entitled data (via the secured access path)
Data steward	Work the exception register; understand treatment applied
Platform operator	Run, monitor, replay and recover the pipeline
Auditor / reviewer	Trace any figure to a rule, a row and a run

B2.3 Operating environment

Python 3.11 or later with an embedded analytical SQL engine for the reference implementation; a dbt port for maintainability; an orchestrator for scheduling. Exact dependency versions are pinned for Python 3.12 and later; Python 3.11 is supported unpinned. Production targets (a partitioned relational warehouse or a cloud warehouse) are described in a reference design and are **not** part of the verified baseline.

B2.4 Design constraints

- All business logic shall be expressed as versioned SQL; orchestration code shall not contain business rules.
- Silver and gold layers shall be reproducible from bronze alone.
- Test and demonstration data shall be generated synthetically and deterministically.

B3. Data requirements

Entity	Source extract	Key	Grain	Layer(s)
Admission	Admissions	AdmissionID	One row per source version (bronze); one per business admission (silver, gold)	Bronze, Silver, Gold
Physician assignment	Physician assignments	DoctorNaturalKey + EffectiveFrom	One row per assignment window (SCD2)	Bronze, Silver, Gold
Billing transaction	Billing extract	TransactionID	One row per latest version of an invoice, credit note or adjustment	Bronze, Silver, Gold
Access grant	Access matrix	User + Branch + Department	Full snapshot per batch; latest snapshot wins	Bronze, Gold
Change event	Change feed	ChangeSequence	One row per change with its availability time	Bronze, Control
Patient, Branch, Department, Date	Derived	Surrogate key	One row per member, plus an Unknown member (-1)	Gold
Reconciliation result	Derived (source files vs bronze)	Run + entity	One row per entity per run; a re-check appends and the latest wins	Control
Configuration parameter	Operator-maintained	Parameter name	One row per tunable threshold; seeded once, never overwritten by a re-run	Control

B4. Functional requirements

B4.1 Ingestion and schema control

ID	Requirement	Trace	Pri.
FR-001	The system shall land each source extract into an append-only bronze table using explicit column lists and record a batch identifier and load timestamp per row.	BRQ-10, 12	M
FR-002	Before each load the system shall compare the incoming file's column names and type families with the last-known-good schema.	BRQ-11	M
FR-003	An added column shall be logged as INFO and shall not alter any downstream business figure until deliberately adopted.	BRQ-11	M
FR-004	A removed column or an unsafe type change shall be logged as BLOCKING and shall skip the load for that entity only; other entities shall load.	BRQ-11	M
FR-005	The trusted schema baseline shall be refreshed only when a load proceeds.	BRQ-11	M

B4.2 Incremental loading and replay

ID	Requirement	Trace	Pri.
FR-010	The system shall support a full load and an incremental load driven by the time each change became available to ETL, not by source modification time.	BRQ-10	M
FR-011	The per-(source, entity) watermark shall advance to the maximum availability time actually processed, and only after the load and transformations succeed.	BRQ-10, 12	M
FR-012	Replaying a batch with an unchanged watermark shall land zero rows.	BRQ-12	M
FR-013	N incremental batches shall produce exactly the same gold state as one full load.	BRQ-12	M
FR-014	Re-running the pipeline after a failure shall produce the state a successful first attempt would have produced.	BRQ-12	M

B4.3 Trusted admissions (silver)

ID	Requirement	Trace	Pri.
FR-020	The system shall retain exactly one row per AdmissionID, selecting the highest source version and breaking ties on the latest modification time. Versions shall never be summed or averaged.	BRQ-01	M
FR-021	The system shall flag: orphan physician key, completed encounter without discharge, discharge before admission, and source-side delete.	BRQ-02	M
FR-022	An admission shall be valid unless its status is Cancelled or Test or it has been deleted at source; a delete shall take effect only once its change event has arrived.	BRQ-01, 10	M
FR-023	An admission with a missing discharge or negative stay shall be quarantined from date-sequenced logic but retained for volume counts.	BRQ-02	M

B4.4 Physician attribution

ID	Requirement	Trace	Pri.
FR-030	The system shall treat assignment end dates as inclusive calendar dates when matching an admission to an assignment.	BRQ-03	M
FR-031	Where two assignments cover the admission date, the system shall select the one with the latest source update and raise a HIGH exception.	BRQ-03	M
FR-032	Where no assignment covers the date, the system shall use the nearest prior assignment (never the current one) and raise a MEDIUM exception.	BRQ-03	M
FR-033	Where no prior assignment exists or the key is unknown, the system shall assign the Unknown member (-1) and, for an unknown key, raise a HIGH exception.	BRQ-02, 03	M

B4.5 Readmission KPI

ID	Requirement	Trace	Pri.
FR-040	An eligible index admission shall be valid, not date-quarantined, and not end in Died or Transfer.	BRQ-04	M
FR-041	For each index the system shall match the first non-Elective admission by the same patient within 30 days after discharge, irrespective of branch.	BRQ-04	M
FR-042	An elective return shall not be counted; an index whose only return is elective shall be flagged.	BRQ-04	M
FR-043	An index discharged within 30 days of the data horizon shall be flagged and excluded from the denominator, and re-evaluated automatically when later data arrives.	BRQ-04	M
FR-044	The KPI shall be exposed at group, branch, department and physician grain from one pair of additive measures (denominator flag, numerator flag).	BRQ-05	M
FR-045	At physician grain the system shall show every physician with a 95% Wilson score interval and a reportable flag, set only when eligible admissions are at or above a configurable minimum (default 25) and the physician is resolved. Reportable physicians shall be listed first and the rows shall still sum to the group total.	BRQ-18	M

B4.6 Revenue reconciliation

ID	Requirement	Trace	Pri.
FR-050	The system shall keep the latest version of each billing transaction.	BRQ-06	M
FR-051	Credit notes and adjustments shall keep their sign and be attributed to the month in which they were raised.	BRQ-06	M
FR-052	The system shall provide a service-period view, a transaction-period view and a bridge whose components sum exactly to service-period revenue.	BRQ-06	M
FR-053	A transaction referencing an unknown admission shall remain in revenue and be flagged and registered as a MEDIUM exception.	BRQ-07	M

B4.7 Data-quality register and outputs

ID	Requirement	Trace	Pri.
FR-060	For every exception the system shall record run, entity, business key, rule code, severity, description and treatment, appending per run.	BRQ-09	M
FR-061	The register shall be derived from the same flags the pipeline acted on.	BRQ-09	M
FR-062	The system shall export the register as a formatted workbook for data stewards.	BRQ-13	S
FR-063	The system should generate a self-contained dashboard of KPIs and exceptions.	BRQ-15	C

B4.8 Security

ID	Requirement	Trace	Pri.
FR-070	A user's visibility shall be the union of all active grants (Enterprise, Branch, Department); inactive grants shall confer nothing.	BRQ-08	M
FR-071	Only the latest access-matrix snapshot shall be used.	BRQ-08	M
FR-072	Grants shall be expanded into a user x branch x department bridge, and a secured access path shall return only entitled rows without duplicating rows for overlapping grants.	BRQ-08	M

B4.9 Monitoring

ID	Requirement	Trace	Pri.
FR-080	The system shall log one row per step per run with status, rows in / out / rejected, timing and any error.	BRQ-12	M
FR-081	A failed step shall be logged and then raised; errors shall never be swallowed.	BRQ-12	M
FR-082	A single alerts view shall return rows for a failed run, blocking schema drift, a HIGH-severity rate above a configurable percentage of valid admissions (default 5%), a source-to-bronze mismatch, a source volume drop and (when enabled) a stale source, and be empty when healthy.	BRQ-12, 17, 21	M

B4.10 Reconciliation and data volume

ID	Requirement	Trace	Pri.
FR-090	After each bronze load the system shall compare, per entity, the source file's row count with the rows landed (this load for a full load, cumulative for an incremental load, the latest snapshot for the access matrix) and record the result.	BRQ-17	M
FR-091	Where an entity has a money column, the system shall also compare the control total of that column between the source file and bronze.	BRQ-17	M
FR-092	The result shall be MATCH, MISMATCH, PENDING (incremental, change feed not yet exhausted) or BLOCKED (halted by breaking schema drift; counted by rows only).	BRQ-17	M
FR-093	A MISMATCH shall raise the alert SOURCE_RECONCILIATION_MISMATCH; PENDING and BLOCKED shall not.	BRQ-17	M
FR-094	The system shall compare each extract's row count with the previous run and raise SOURCE_VOLUME_DROP when it shrinks by more than a configurable percentage (default 30%).	BRQ-21	M
FR-095	The system shall report hours since each source system's latest change and, when a configurable threshold is greater than zero, raise STALE_SOURCE above it (default 0 = disabled).	BRQ-21	S

B4.11 Configuration and notification

ID	Requirement	Trace	Pri.
FR-100	Alert and reporting thresholds shall be stored in a configuration table, seeded once and never overwritten by a re-run.	BRQ-20	S
FR-101	The views shall read the configuration at query time, so a changed value takes effect without a software release.	BRQ-20	S
FR-102	On request the system shall post the current alerts to a configured webhook as one message.	BRQ-19	S
FR-103	A run with no alerts shall send nothing.	BRQ-19	S
FR-104	A failed or non-successful delivery shall be logged and reported but shall not alter or fail the pipeline run.	BRQ-19	S
FR-105	With no webhook configured, the system shall print the alerts and report that nothing was sent.	BRQ-19	S

B5. External interfaces

Interface	Direction	Format	Notes
Source extracts (5)	In	CSV, header row, UTF-8	Agreed column set; version and modification timestamp present
Mart exports	Out	CSV	Gold facts, KPI views, control tables
Exception register	Out	XLSX	Formatted for stewards
Query access	Out	SQL	Gold schema, KPI views, secured access path
Scheduler	In/Out	DAG tasks	Land, transform, quality gate, dbt build, publish
Alert webhook	Out	HTTPS POST, JSON (Slack-compatible)	One message per run with alerts; none when healthy

B6. Non-functional requirements

ID	Category	Requirement	Measure	Pri.
NFR-01	Performance	A full load of about 1 million source rows shall complete within 5 minutes on a 2-vCPU host.	Benchmark timing	M
NFR-02	Reliability	Re-runs shall be idempotent; gold shall never be left half-updated by a retry.	Two runs give identical tables	M
NFR-03	Data integrity	Facts shall have a unique grain key; every foreign key shall resolve (to a member or the Unknown member).	Uniqueness and relationship tests	M
NFR-04	Auditability	Any reported figure shall be traceable to a numbered rule, a row and a run.	Rule IDs in SQL headers; run log	M
NFR-05	Maintainability	Business logic shall live in SQL; a second implementation (dbt) shall be proven equivalent by tests.	Row-for-row equivalence	S
NFR-06	Testability	Rules shall be verified against an independent implementation and a seeded-defect manifest; rule sensitivity shall be measured by mutation.	Oracle, recall / precision, mutation	M
NFR-07	Security	Non-production data shall be synthetic; production deployment shall add masking, encryption and access auditing.	Review gate	M
NFR-08	Portability	A reference design shall exist for a partitioned relational and a cloud warehouse; dialect differences shall be documented.	Reference DDL	S
NFR-09	Scalability	The design shall define incremental MERGE-based loading for volumes beyond tens of millions of rows.	Reference design	S
NFR-10	Operability	A runbook shall describe diagnosis and recovery for each failure point.	Runbook review	M
NFR-11	Reproducibility	Dependency versions shall be pinned exactly for the verified environment and shown to reproduce it.	Fresh install equals the pin file; no broken requirements	S
NFR-12	Code quality	Code shall pass the configured static analysis in CI.	Linter clean	S
NFR-13	Alert coverage	Every alert type shall have an automated test that makes it fire, and adding an alert without one shall fail the build.	Failure-path tests plus a tripwire test	M

B7. Business-rule catalogue

Rule	Statement	Realised by
BR-1	A valid admission is the latest version of an encounter, excluding Cancelled, Test and source-deleted encounters.	FR-020, 022
BR-2	Latest version wins; versions are never summed or averaged.	FR-020, 050
BR-3	Attribute activity to the assignment effective on the event date.	FR-030 to 033
BR-4	Index eligibility: valid, not quarantined, not Died or Transfer.	FR-040
BR-5	Exclude planned (Elective) returns; match the first unplanned return in 30 days.	FR-041, 042
BR-6	Readmission search is not restricted to the index branch.	FR-041
BR-7	Exclude indexes without a full 30-day observation window from the denominator.	FR-043
BR-8	Credit notes and adjustments keep their sign and belong to the month raised.	FR-051

B8. Exception-rule requirements

Rule code	Severity	Trigger	Required treatment
`DUP_SOURCE_VERSION`	MEDIUM	More than one version of a key	Keep latest; retain history
`NON_INPATIENT_ENCOUNTER`	LOW	Status Cancelled or Test	Exclude from valid; retain
`ORPHAN_DOCTOR_FK`	HIGH	Physician key unknown	Unknown member; retain
`NEGATIVE_LOS / MISSING_DISCHARGE`	HIGH	Impossible or missing discharge	Quarantine from date logic
`SOURCE_DELETED`	MEDIUM	Delete event after latest version	Soft delete
`SCD_OVERLAP / SCD_COVERAGE_GAP`	HIGH / MEDIUM	Overlapping or missing assignment	Latest update / nearest prior
`ORPHAN_BILLING_ADMISSION`	MEDIUM	Invoice with unknown admission	Keep in revenue; flag
`ILLOGICAL_ADMIT_AFTER_DEATH`	HIGH	Admission after a Died disposition	Retain both; identity review
`INSUFFICIENT_OBSERVATION_WINDOW`	LOW	Index within 30 days of horizon	Exclude from denominator

B8.1 Alert conditions

Alert	Condition	Default threshold
`PIPELINE_FAILED`	A step of the latest run failed	-
`BLOCKING_SCHEMA_DRIFT`	Breaking drift halted an entity	-
`HIGH_SEVERITY_DQ_RATE`	HIGH exceptions as a share of valid admissions	5% (configurable)
`SOURCE_RECONCILIATION_MISMATCH`	Bronze differs from the source file	-
`SOURCE_VOLUME_DROP`	Extract shrank versus the previous run	30% (configurable)
`STALE_SOURCE`	No change from a source for too long	Off (configurable hours)

B9. Verification and traceability

Each requirement is verified by test (T), inspection (I) or demonstration (D). **Status** reflects the implemented reference platform: **Verified** = automated test passing; **Design only** = specified and reviewed but not executed in a live environment.

Requirement (s)	Implemented in	Verification	Method	Status
FR-001 to 005	`pipeline/schema_guard.py`, `sql/00, 01`	`test_schema_guard.py` (5 tests)	T	Verified
FR-010 to 014	`run_pipeline.py`, `sql/10`	`test_incremental.py` (8); idempotency test	T, D	Verified
FR-020 to 023	`sql/03`	Oracle row-by-row comparison; grain and no-drop tests	T	Verified
FR-030 to 033	`sql/04`	Oracle comparison; edge case: assignment boundary day	T	Verified
FR-040 to 044	`sql/07`	Oracle per index row; 30-day boundary to the minute; grain reconciliation	T	Verified
FR-045	`sql/07`	Interval equals an independent reference for every row; flag follows config; UNKNOWN never reportable; rows sum to the group total	T	Verified
FR-050 to 053	`sql/08`	Oracle monthly revenue; bridge sums	T	Verified
FR-060 to 062	`sql/10`, `export_dq_register.py`	Planted-defect recall and precision	T	Verified
FR-063	`build_dashboard.py`	Generated artefact reviewed	D	Verified
FR-070 to 072	`sql/09`	Per-user visibility vs oracle; revoked-grant mutation	T	Verified
FR-080 to 082	`sql/11`, `run_pipeline.py`	Run-log completeness; every alert type fires on an injected fault and stays silent when healthy (`test_enhancements.py`)	T	Verified
FR-090 to 095	`run_pipeline.py`, `sql/11`	Clean run MATCH; dropped rows; same count but altered amount; re-run; incremental PENDING then MATCH; blocked entity; half-empty extract; stale source	T	Verified
FR-100 to 105	`sql/02`, `pipeline/notify.py`	Stub sender, a real local HTTP server and an unreachable URL; silent when healthy. A real chat or paging tool was not tested	T	Partly verified
Scheduler (B5)	`airflow/dags/`	`test_dag.py` (structural)	I	Design only
NFR-01	`make bench`	~8.4 s for ~1.06M rows on 1 vCPU / 3 GB (limit 5 minutes)	T	Verified
NFR-05	`dbt/`	dbt build: 13 models, 32 tests, 4 equivalence tests	T	Verified
NFR-06	`tests/oracle.py`, `scripts/mutation_check.py`	104 tests; 12/12 mutations killed	T	Verified
NFR-08, 09	`sql/12`	Reference DDL reviewed; never executed	I	Design only
NFR-10	`docs/runbook.md`	Runbook review	I	Verified
NFR-11	`constraints.txt`	Fresh install with the pins equals the pin file; `pip check` clean (Python 3.12). Python 3.11 not run	T	Verified (3.12)
NFR-12	`pyproject.toml`, CI lint job	ruff clean	T	Verified
NFR-13	`tests/test_enhancements.py`	A failure-path test for each alert type and a tripwire that fails if an alert has no test	T	Verified

Traceability rule
Every Must-priority business requirement (BRQ) maps to at least one functional requirement (FR) and every FR to at least one verification. Gaps are listed in B10 rather than hidden.

B10. Open issues and items to be decided

#	Item	Owner	Needed by
OI-1	Confirm clinical interpretation: planned-return exclusion applies to the candidate readmission; transfers excluded from index	[Clinical Quality Lead]	[date]
OI-2	Replace 'as-of = latest modification time' with the extract timestamp from the source contract	[Data Engineering]	[date]
OI-3	Decide the production warehouse and execute the reference DDL in CI	[Head of Data and Analytics]	[date]
OI-4	Define PHI controls: masking, access audit, retention	[Information Security Officer]	[date]
OI-5	Tune alert thresholds in the configuration table from historical run data (adjustable without a release)	[Data Engineering]	[date]
OI-6	Run the orchestrator DAG in a live environment; bound loads by data interval	[Platform Operator]	[date]
OI-7	CLOSED in v1.0: failure-path tests added for every alert type, with a tripwire for new alerts	[Data Engineering]	Closed
OI-8	Audit changes to the configuration table (who, when, previous value)	[Data Engineering]	[date]
OI-9	Verify alert delivery against the real chat or paging tool and add de-duplication	[Platform Operator]	[date]
OI-10	Confirm the Python 3.11 compatibility leg in CI (the exact pins require 3.12 or later)	[Data Engineering]	[date]
OI-11	Decide stable surrogate keys (for example hash keys) together with the dbt equivalence tests	[Head of Data and Analytics]	[date]
OI-12	Add case-mix attributes before provider comparisons are used for decisions	[Clinical Quality Lead]	[date]

B11. Glossary

Term	Definition
Index admission	The admission from which a 30-day readmission window is measured
Unplanned readmission	A non-Elective admission within 30 days after an index discharge
SCD2	Slowly changing dimension type 2: history kept as validity windows
Watermark	High-water mark of data already processed per source and entity
Quarantine	Keep a row for counts and audit but exclude it from logic it would corrupt
Unknown member	Dimension row with key -1 used when a lookup fails
RLS	Row-level security
HIS	Hospital Information System
MoSCoW	Must / Should / Could / Won't prioritisation
Reconciliation	Checking that what was loaded equals what the source sent
Control total	A sum of a money column compared between source and bronze, to catch altered values a row count cannot
Wilson score interval	A confidence interval for a proportion that stays sensible at small samples